

4. (a) Here, $x = 12$ and $y = 36$
 $\therefore$ Time taken by $B = \frac{xy}{y-x} = \frac{12 \times 36}{36-12}$

$$= \frac{12 \times 36}{24} = 18 \text{ days}$$
 [Rule 3]

5. (c) $\therefore$ One day work of Ram $= \frac{1}{6}$
 $\therefore$ One day work of Shyam $= \frac{1}{12}$
Hence, one day work of, Ram and Shyam

$$= \frac{1}{6} + \frac{1}{12} = \frac{2+1}{12} = \frac{3}{12} = \frac{1}{4}$$

 $\therefore$ Two day's work $= \frac{1}{2}$

Thus, if they work together for 2 day's, then half of the work will be complete.

6. (b) Here, $x = 8$, $y = 12$, $z = 12$
Now, time taken by $(A + B + C)$

$$= \frac{2xyz}{xy + yz + zx}$$
 [Rule 4]

$$\Rightarrow 6 = \frac{2 \times 8 \times 12 \times 12}{8 \times 12 + 12 \times 12 + 8 \times 12}$$

$$\Rightarrow 96 + 20z = 32z$$

$$\Rightarrow 96 = 12z$$

$$\Rightarrow z = 8$$

So, A and C can do the work in 8 days.

7. (b) One day's work of $A = \frac{1}{8}$
One day's work of $B = \frac{1}{12}$
3 day's work of $A = \frac{3}{8}$
Remaining work of $A = 1 - \frac{3}{8}$

$$= \frac{5}{8}$$

One day's work of A and B together

$$= \frac{1}{8} + \frac{1}{12} = \frac{3+2}{24} = \frac{5}{24}$$

Number of days to finish the work

$$= \frac{5}{\frac{5}{24}} = 24 \text{ days}$$

8. (d) 1 min work of first tap $= \frac{1}{5}$
1 min work of second tap $= \frac{1}{7}$
and 1 min work of third $= \frac{1}{3}$

$$\therefore 1 \text{ min work of all taps} = \frac{1}{5} + \frac{1}{7} + \frac{1}{3}$$

$$= \frac{21+15+35}{105} = \frac{1}{105}$$

Hence, tap will be filled in 105 min, if they work together.

9. (b) Here, $a = 40$ days, $b = 50$ days,
 $x = 20$ and $T = ?$
 $\therefore$ Required time $= \frac{(b-x)a}{a+b}$

$$= \frac{(50-20) \times 40}{(40+50)} = \frac{30 \times 40}{90}$$

$$= \frac{40}{3} = 13\frac{1}{3} \text{ days}$$

10. (b) Here, $x = 2$, $y = 4$ and $z = \frac{12}{5}$
 $\therefore$ Time taken by $(P + Q + R)$

$$= \frac{2xyz}{xy + yz + zx}$$
 [Rule 4]

$$= \frac{2 \times 2 \times 4 \times \frac{12}{5}}{2 \times 4 + 4 \times \frac{12}{5} + \frac{12}{5} \times 2}$$

$$= \frac{192}{40 + 48 + 24}$$

$$= \frac{192}{112} = \frac{12}{7} \text{ days}$$

Now, P one days work

$$= \frac{7}{12} - \frac{1}{4} = \frac{7-3}{12}$$

$$= \frac{4}{12} = \frac{1}{3}$$

Hence, P alone can do in 3 days.

11. (d) Using the formula $\frac{M_1 D_1}{W_1} = \frac{M_2 D_2}{W_2}$

12. (a) By given condition,
 $n \times 30 = n \times 10 + (n + 50) \times 16$
 $\Rightarrow 20n = 16n + 800$
 $\therefore n = \frac{800}{4} = 200$

13. (c) Here, $x = 12$, $y = 16$ and $t = 9$
Two pipes A and B can fill a tank in x h and y h, respectively. If both the pipes ax opened simultaneously, then the time after which B should be closed, so that the tank is filled in
 t h $= [y(1 - t/x)]^n$
Required time after which B should be closed $= y\left(1 - \frac{t}{x}\right) = 16\left(1 - \frac{9}{12}\right)$

$$= 16 \times \frac{3}{12} = 4 \text{ min}$$

14. (a) Let the boy completes the work in x days.

According to the condition,

$$\frac{1}{7} + \frac{1}{8} + \frac{1}{x} = \frac{1}{3} \Rightarrow \frac{1}{x} = \frac{1}{3} - \frac{1}{8} - \frac{1}{7}$$

$$\Rightarrow \frac{1}{x} = \frac{56-21-24}{168} = \frac{11}{168}$$

$$\therefore x = \frac{168}{11} \text{ days}$$

So, money is to be shared in the ratio

$$\frac{1}{7} : \frac{1}{8} : \frac{11}{168} \text{ or } 24 : 21 : 11$$

Thus,

$$A's \text{ amount} = \frac{24}{56} \times 1400 = ₹ 600$$

$$B's \text{ amount} = \frac{21}{56} \times 1400 = ₹ 525$$

$$\text{Boy's amount} = \frac{11}{56} \times 1400 = ₹ 275$$

15. (d) A 's work in one day $= \frac{1}{8}$, B 's work in one day $= \frac{1}{10}$

$$C's \text{ work in one day} = \frac{1}{15}$$

$$(A+B)'s \text{ work in one day} = \frac{1}{8} + \frac{1}{10} = \frac{5+4}{40} = \frac{9}{40}$$

$$(A+B)'s \text{ work in two days} = \frac{2 \times 9}{40} = \frac{9}{20}$$

$$\text{Remaining work} = 1 - \frac{9}{20} = \frac{11}{20}$$

$$(B+C)'s \text{ work in one day} = \frac{1}{10} + \frac{1}{15} = \frac{3+2}{30} = \frac{5}{30} = \frac{1}{6}$$

Since, $(B+C)$ complete the work in 6 days.

$$\therefore \frac{11}{20} \text{ work will be completed in}$$

$$6 \times \frac{11}{20} = \frac{11 \times 3}{10} = \frac{33}{10} \text{ days}$$

$$\therefore \text{Total number of days} = 2 + \frac{33}{10} = \frac{20+33}{10} = \frac{53}{10} \text{ days}$$

16. (d) 1 day work of $A = \frac{1}{x}$

$$1 \text{ day work of } B = \frac{1}{3x}$$

$$\therefore 1 \text{ day work of both } A \text{ and } B$$

$$= \frac{1}{x} + \frac{1}{3x} = \frac{4}{3x}$$

$$\text{given, one day work of both } A \text{ and } B = \frac{1}{12}$$

$$\Rightarrow \frac{4}{3x} = \frac{1}{12} \Rightarrow 3x = 48 \Rightarrow x = 16$$

Hence, the value of x is 16.

- 17. (a)** Given, $6M + 8B = 10$ days ... (i)
and $26M + 48B = 2$ days ... (ii)
 $15M + 20B = ?$

Here, $M_1 D_1 = M_2 D_2$
 $\Rightarrow (6M + 8B) \times 10 = (26M + 48B) \times 2$
 $\Rightarrow 60M + 80B = 52M + 96B$
 $\Rightarrow 8M = 16B$
 $\therefore M = 2B$
 Then, $15M + 20B = 15 \times 2B + 20B$
 $= 30B + 20B = 50B$

On putting the value of M in Eq. (i), we get
 $\Rightarrow 6 \times 2B + 8B = 10$ days
 $\Rightarrow 12B + 8B = 10$ days
 $\therefore 20$ boys finish the work in 10 days
 One boy finish the work in
 10×20 days = 200 days
 and 50 boys finish the work in $\frac{200}{50}$ days
 $= 4$ days

- 18. (b)** Part filled by first tap in 1h = 1
 Part filled by second tap in 1h = $\frac{1}{2}$
 Part filled by third tap in 1h = $\frac{1}{3}$
 Part filled by fourth tap in 1h = $\frac{1}{6}$
 Total tank filled by all taps in 1h
 $= 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{6}$
 $\therefore$ Required time = $\frac{1}{1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{6}}$
 $= \frac{6}{6 + 3 + 2 + 1} = \frac{6}{12} \text{ h} = \frac{6}{12} \times 60 \text{ min}$
 $= 30 \text{ min}$

- 19. (c)** Given, $M_1 = 76, D_1 = 33$
 Let number of ladies who did not report for the work = x
 By given condition,
 $M_2 = 76 - x$ and $D_2 = 44$
 We know that,
 $M_1 D_1 = M_2 D_2$ [Rule 5]
 $\therefore 76 \times 33 = (76 - x) \times 44$
 $\Rightarrow 76 - x = \frac{76 \times 3}{4} = 19 \times 3 \Rightarrow x = 19$

Hence, the number of ladies is 19.

- 20. (b)** Refer to Question No 11
 [Using Rule 5]

- 21. (c)** Work done by Ravi in one hour = $\frac{1}{8}$

Work done by Sneha in one hour = $\frac{1}{12}$

Total work done by them in 2 h
 $= \frac{1}{8} + \frac{1}{12} = \frac{5}{24}$

Work done is 4 pairs of hours
 $= \frac{5}{24} \times 4 = \frac{5}{6}$

Remaining work after 8 hours
 $= 1 - \frac{5}{6} = \frac{1}{6}$

Now, it's Ravi's turn

So, work left after 9 hours = $\frac{1}{6} - \frac{1}{8} = \frac{1}{24}$

$\frac{1}{24}$ work will be done by sneha in

$$12 \times \frac{1}{24} \text{ h} = 30 \text{ min}$$

Total time taken = 9 h and 30 min

Required time = 6 : 30 pm

- 22. (d)** I. 18 men can earn in 5 days = ₹ 1440

1 man can earn in 1 day = ₹ $\frac{1440}{18 \times 5}$

$\therefore$ 10 men can earn in 6 days

$$= \frac{1440}{18 \times 5} \times 6 \times 10$$

$$= ₹ 960 \neq ₹ 1280$$

II. 16 men can earn in 7 days = ₹ 1120

1 man can earn in 1 day = $\frac{1120}{16 \times 7}$

$\therefore$ 21 men can earn in 4 days

$$= \frac{1120}{16 \times 7} \times 21 \times 4 = ₹ 840 \neq ₹ 800$$

So, neither statement I nor II is correct.

- 23. (a)** Here, $a_1 = 14, b_1 = 12, x = 4,$

$$a_2 = 8, b_2 = 16 \text{ and } y = 5$$

$$\frac{\text{One day work of 1 man}}{\text{One day work of 1 boy}} = \frac{(yb_2 - xb_1)}{(xa_1 - ya_2)}$$

$$= \frac{5 \times 16 - 4 \times 12}{4 \times 14 - 5 \times 8} = 2$$

- 24. (b)** Refer to Question No 23.

- 25. (b)** Using the formula, $M_1 D_1 = M_2 D_2$

- 26. (d)** Here, $x = 10, y = 15, z = 12.$

$\therefore$ Time taken by $(A + B + C)$

$$= \frac{2xyz}{xy + yz + zx} \quad [\text{Rule 4}]$$

$$= \frac{2 \times 10 \times 15 \times 12}{10 \times 15 + 15 \times 12 + 12 \times 10}$$

$$= \frac{3600}{450} = 8 \text{ h}$$

So, 1 h work of $B = \frac{1}{8} - \frac{1}{12}$
 $= \frac{3-2}{24} = \frac{1}{24}$

Hence, B alone can do in 24 h.

- 27. (b)** Here, $x = 25$ days and $y = 25$

$\therefore$ Required days = $\frac{100x}{100 + y} = \frac{100 \times 25}{100 + 25}$
 $= \frac{2500}{125} = 20$ days

- 28. (c)** Mason work for 1 h = $\frac{1}{12}$

Mason Work for 6 h = $\frac{6}{12} = \frac{1}{2}$

Work left = $1 - \frac{1}{2} = \frac{1}{2}$

Now, let the boy can finish the work in x h.

Then, their 1 h work = $\frac{1}{12} + \frac{1}{x} = \frac{x + 12}{12x}$

$\therefore \frac{x + 12}{12x} \times 5 = \frac{1}{2} \Rightarrow \frac{5x + 60}{12x} = \frac{1}{2}$

$\Rightarrow 10x + 120 = 12x \Rightarrow 120 = 2x$

$\therefore x = 60$ h

- 29. (c)** X 's one day work = $\frac{1}{12}$ and $(X + Y)$'s

one day work = $\frac{3}{20}$

$\therefore Y$'s one day work

$$= \frac{3}{20} - \frac{1}{12} = \frac{4}{60} = \frac{1}{15}$$

$\therefore$ Number of day's taken by Y to complete the work = 15 days

- 30. (c)** Part filled by pipe A in 1 min = $\frac{1}{10}$

and part emptied by pipe B in 1 min = $\frac{1}{15}$

$\therefore$ Total tank filled in minutes

$$= \frac{1}{10} - \frac{1}{15} = \frac{3-2}{30} = \frac{1}{30}$$

Hence, the tank will be filled in 30 min.

- 31. (d)** Here, $m = 4, n = 6, a = 50$ and $x = 2, y = 9$

$\therefore$ Required days = $\frac{1}{\frac{x}{m \times a} + \frac{y}{n \times a}}$

$$= \frac{1}{\frac{2}{4 \times 50} + \frac{9}{6 \times 50}} \quad [\text{Rule 6}]$$

$$= \frac{1}{\frac{1}{100} + \frac{3}{100}} = \frac{1}{\frac{4}{100}} = \frac{100}{4} = 25 \text{ days}$$

32. (a) Here, $x = 15$ days, $y = 20$ days and $z = 25$ days and $K = ₹4700$

$$\begin{aligned}\text{Share of } C &= ₹ \left(\frac{kxy}{xy + yz + zx} \right) \\ &= ₹ \frac{4700 \times 15 \times 20}{15 \times 20 + 20 \times 25 + 25 \times 15} \\ &= ₹ \frac{4700 \times 15 \times 20}{1175} = ₹ 1200\end{aligned}$$

33. (a) Refer to Question No 11

[Using Rule 5]

34. (b) Refer to Question No 22

35. (b) $\therefore$ 14 days work of 2 men and 1 women
 $= 8$ days work of 4 women and 2 men
 $\Rightarrow$ 1 day work of 28 men and 14 women
 $=$ 1 day work of 32 women and 16 men
 $\Rightarrow$ 28 men + 14 women
 $=$ 32 women + 16 men
 $\Rightarrow 28m - 16m = 32w - 14w$
 $\Rightarrow 12m = 18w \Rightarrow \frac{m}{w} = \frac{18}{12} = \frac{3}{2}$

So, efficiency of 1 man and 1 woman is 3 : 2.

So, their wages must be in the same ratio

$$\therefore \frac{90}{x} = \frac{3}{2}$$

[where, x = wages per day of a woman]

$$\therefore x = \frac{90 \times 2}{3} = ₹ 60$$

36. (b) A's one day work = $\frac{1}{4}$

$$\text{and } B\text{'s one day work} = \frac{1}{12}$$

One day work of A and B together

$$= \frac{1}{4} + \frac{1}{12} = \frac{3+1}{12} = \frac{4}{12} = \frac{1}{3}$$

Days required by A and B together to do the work

$$= \frac{1}{\text{One day work of A and B together}} = 3 \text{ days}$$

37. (a) Work done by A in one day = $\frac{1}{8}$

$$\text{Work done by B in one day} = \frac{1}{12}$$

$$\text{and work done by C in one day} = \frac{1}{15}$$

Let the work will be completed in x days.

$$\text{Then, } \frac{2}{8} + \frac{x}{12} + \frac{x-2}{15} = 1$$

$$\Rightarrow \frac{1}{4} + \frac{x}{12} + \frac{x-2}{15} = 1$$

$$\Rightarrow \frac{15 + 5x + 4(x-2)}{60} = 1$$

$$\Rightarrow 15 + 5x + 4x - 8 = 60 \Rightarrow 9x + 7 = 60$$

$$\Rightarrow 9x = 53 \Rightarrow x = \frac{53}{9} = 5\frac{8}{9}$$

Hence, the work will completed in $5\frac{8}{9}$ days.

38. (b) Let efficiency of A be x days.

Then, efficiency of B = $3x$ days.

$$\therefore B\text{'s one day work} = \frac{1}{3x}$$

$$\text{and } A\text{'s one day work} = \frac{1}{x}$$

Again, let time taken by B = t days

Then, time taken by A = $(t - 40)$ days

$$\text{Now, } \frac{t}{3x} = \frac{t-40}{x} \Rightarrow t = 3(t-40)$$

$$\Rightarrow 3t - t = 120, t = \frac{120}{2} = 60 \text{ days}$$

So, B complete the work in 60 days and A complete the work in 20 days.

(A + B)'s 1 day work

$$= \frac{1}{60} + \frac{1}{20} = \frac{4}{60} = \frac{1}{15}$$

Hence, they will complete the work in 15 days when they work together.

39. (a) Let efficiency of P be x days.

Then, efficiency of Q = $2x$ days

$$\therefore Q\text{'s one day work} = \frac{1}{2x}$$

$$\text{and } P\text{'s one day work} = \frac{1}{x}$$

Now, (P + Q)'s one day work

$$= \frac{1}{x} + \frac{1}{2x} = \frac{3}{2x}$$

$\therefore$ (P + Q) will complete the whole work in $\frac{2x}{3}$ days.

According to the question,

R will complete this work = $2x$ days

$$\therefore R\text{'s one day work} = \frac{1}{2x}$$

$$\begin{aligned}\text{Required ratio} &= \frac{1}{x} : \frac{1}{2x} : \frac{1}{2x} = 1 : \frac{1}{2} : \frac{1}{2} \\ &= 2 : 1 : 1\end{aligned}$$

40. (d) Part of tank filled by A in 1 min = $\frac{1}{10}$

$$\text{Part of tank filled by B in 1 min} = \frac{1}{20}$$

$$\begin{aligned}\text{Part of tank emptied by leakage in 1 min} \\ &= \frac{1}{40}\end{aligned}$$

$$\text{Part of tank filled by both taps and leakage is } \left(\frac{1}{10} + \frac{1}{20} - \frac{1}{40} \right) = \frac{1}{8}$$

Hence, total time taken to fill the tank is 8 min.

41. (c) Here, $M_1 = 4, T_1 = 4, W_1 = 4, D_1 = 4$ and $M_2 = 2, T_2 = 2, D_2 = 2, W_2 = ?$

Using the formula,

$$\frac{M_1 \times T_1 \times D_1}{W_1} = \frac{M_2 \times T_2 \times D_2}{W_2} \quad [\text{Rule 5}]$$

$$\Rightarrow \frac{4 \times 4 \times 4}{4} = \frac{2 \times 2 \times 2}{W_2}$$

$$\Rightarrow W_2 = \frac{8}{16} = \frac{1}{2}$$

42. (d) m persons paint a house in d days.

$\therefore$ 1 person paints a house in $(m \times d)$ days.

and $m + 2$ persons paint a house in

$$\left(\frac{md}{m+2} \right) \text{ days.}$$

43. (a) Since, two pipes A and B fill a tank in 60 min and 75 min, respectively.

$\therefore$ Part of tank filled by pipe A in 1 min

$$= \frac{1}{60}$$

and part of tank filled by pipe B in 1 min

$$= \frac{1}{75}$$

Now, part of tank filled by A and B together in 1 min

$$= \frac{1}{60} + \frac{1}{75} = \frac{9}{300} = \frac{3}{100}$$

Let part of tank emptied by pipe C in 1 min = $\frac{1}{C}$.

So, net part of tank filled by pipes A, B and C together in 1 min

$$= \frac{3}{100} - \frac{1}{C} = \frac{1}{50} \quad [\text{Given}]$$

$$\Rightarrow \frac{1}{C} = \frac{3}{100} - \frac{1}{50} = \frac{1}{100}$$

$\therefore C = 100$

Hence, the time taken by pipe C to empty the tank is 100 min.

08

PERCENTAGE

Generally (2-3) questions have been asked from this chapter. Generally questions which are asked from this chapter are tricky but you can easily solve them by using the short-tricks formulae in very less time.



PER CENT

Per cent is a fraction whose denominator is 100 and numerator of the fraction is called the rate per cent. Per cent is denoted by the symbol '%'.
↓

Important Rules and Formulae

Rule 1 For expressing $a\%$ as a fraction, we can write $a\% = \frac{a}{100}$, i.e. divide 'a' by 100 and reduce it to the lowest form, e.g. $25\% = \frac{25}{100} = \frac{1}{4}$

Rule 2 For expressing a fraction $\frac{a}{b}$ as a per cent, we can write $\frac{a}{b} = \left(\frac{a}{b} \times 100\right)\%$

e.g. $\frac{2}{3} = \frac{2}{3} \times 100\% = \frac{200}{3}\%$

Rule 3 To find how much per cent one quantity is of another quantity, we can write

Required percentage = $\frac{\text{The quantity to be expressed in per cent}}{\text{2nd quantity (in respect of which the per cent has to be obtained)}} \times 100\%$

EXAMPLE 1. 20g is what per cent of 1000g?

a. 2%

b. 5%

c. 7%

d. 9%

Sol. a. Required percentage = $\frac{20}{1000} \times 100\% = 2\%$

Rule 4 Percentage increase or decrease in a quantity when it increases or decreases by some value.

Percentage = $\frac{\text{Increase or decrease in the value}}{\text{Initial value}} \times 100\%$

EXAMPLE 2. A person's salary has increased from ₹ 7200 to ₹ 8100. What is the percentage increase in his salary? ✎ 2013 II

- a. 25% b. 18% c. $16\frac{2}{3}\%$ d. $12\frac{1}{2}\%$

Sol. d. Percentage increase in salary = $\frac{8100-7200}{7200} \times 100\%$
 $= \frac{900}{7200} \times 100\% = 12.5\% = 12\frac{1}{2}\%$

(Rule 5) If A is $x\%$ more than ' B ', then B would be $\left[\frac{x}{(100+x)} \times 100 \right]\%$ less than ' A '.

(Rule 6) If A is $x\%$ less than ' B ', then B would be $\left[\frac{x}{(100-x)} \times 100 \right]\%$ more than A .

EXAMPLE 3. Raj get 10% less marks than Rohit in an examination. What percentage of marks does Rohit gets more than Raj?

- a. $11\frac{1}{6}\%$ b. $11\frac{1}{9}\%$ c. $12\frac{1}{9}\%$ d. None of these

Sol. b. Given, $x = 10\%$
 $\therefore$ Rohit more percentage than Raj = $\frac{10}{100-10} \times 100$
 $= \frac{10 \times 100}{90} = 11\frac{1}{9}\%$

(Rule 7) If the price of a commodity is increased (or decreased) by $x\%$, then the decrease (or increase) in consumption, so as not to increase or (decrease) the expenditure is $\left(\frac{x}{100 \pm x} \times 100 \right)\%$.

EXAMPLE 4. If the price of the cooking gas rises by 15%, by what per cent should a family reduce its consumption so as not to exceed the budget on cooking gas?

- a. $12\frac{1}{23}\%$ b. $13\frac{1}{23}\%$ c. $14\frac{1}{23}\%$ d. None of these

Sol. b. Let initial price of cooking gas be ₹ 100.

Price after increase = ₹ 115

On ₹ 115 he should reduce ₹ 15 on ₹ 100, he should reduce

$$= \frac{15}{115} \times 100 = 13\frac{1}{23}\%$$

Shortcut Method

Here, $x = 15\%$

$$\therefore \text{Reduction in consumption} = \left(\frac{x}{100+x} \times 100 \right)\% \\ = \frac{15}{115} \times 100\% = 13\frac{1}{23}\%$$

(Rule 8) If the value of an object is first changed (increased or decreased) by $x\%$ and then changed (increased or decreased) by $y\%$, then

$$\text{Net effect} = \left(\pm x \pm y + \frac{(\pm x)(\pm y)}{100} \right)\%$$

Here, + sign is used in case of increment and - sign is used in case of decrement.

EXAMPLE 5 The price of an article is first increased by 20% and later on the price were decreased by 25% due to reduction in sales. Find the net percentage change in final price of article.

- a. 20% b. 10% c. 30% d. 15%

Sol. b. Here, $a = 20\%$, $b = 25\%$

$$\text{So, required percentage} = \frac{10}{100} \times 100 = 10\%$$

$$\begin{aligned} \text{Required change} &= \left[(\pm a) + (\pm b) + \frac{(\pm a)(\pm b)}{100} \right]\% \\ &= \left[20 - 25 + \frac{20 \times (-25)}{100} \right]\% \begin{cases} \text{positive sign for increase} \\ \text{negative sign for decrease} \end{cases} \\ &= [-5 - 5]\% \\ &= -10\% \end{aligned}$$

$\therefore$ Net percentage change is a decrease of 10 % because final result is negative.

(Rule 9) If the population of a town is P and it increases (or decreases) at the rate of $R\%$ per annum, then

$$(i) \text{ Population after } n \text{ yr} = P \left(1 \pm \frac{R}{100} \right)^n$$

$$(ii) \text{ Population, } n \text{ yr ago} = \frac{P}{\left(1 \pm \frac{R}{100} \right)^n}$$

EXAMPLE 6 The population of a town is 352800. If it increases at the rate of 5% per annum, then what will be its population 2 yr hence, also find the population 2 yr ago.

- a. 315000 b. 316500 c. 200045 d. 320000

Sol. d. Given that, $p = 352800$, $R = 5\%$ and $n = 2$

According to the formula,

$$\begin{aligned} \text{Population after 2 yr} &= P \left(1 + \frac{R}{100} \right)^n = 352800 \times \left(1 + \frac{5}{100} \right)^2 \\ &= 352800 \times \left(\frac{100+5}{100} \right)^2 = 352800 \times \left(\frac{21}{20} \times \frac{21}{20} \right) = 388962 \end{aligned}$$

$$\begin{aligned} \text{Population 2 yr ago} &= \frac{P}{\left(1 + \frac{R}{100} \right)^n} = \frac{352800}{\left(1 + \frac{5}{100} \right)^2} = 352800 \times \frac{20}{21} \times \frac{20}{21} = 320000 \end{aligned}$$